

Dohyun (Eric) Kim

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PROFILE

I'm a Penultimate Computer Systems Engineering (Hons) student at the University of Auckland, passionate about the hardware-software boundary. I bring a strong project portfolio, AWS certifications, and club executive experience.

EDUCATION

University of Auckland - Auckland CBD, Auckland

Expected Graduation **Nov 2027**

- **Degree:** Bachelor of Engineering (Hons) in Computer Systems Engineering
- **Concentration:** Embedded Systems & Advanced Software Skills

PROJECTS

Sentiment PULSE - AWS | Individual Project | 2026

- Built a **serverless sentiment analysis pipeline** using AWS **Kinesis, Lambda, Comprehend, and DynamoDB** to demonstrate end-to-end cloud and AI engineering on AWS.
- Developed an event-driven NLP architecture with a live React dashboard using Kinesis-triggered Lambda and Comprehend sentiment classification **to showcase real-time data ingestion**, inference, and visualisation.

“Winnie the bot” - [3rd Place] ECSE Design Competition | Team Project | 2025

- Built **embedded hardware** for an AI-powered interactive robot using dual ATmega328P microcontrollers, servos, an AI camera, and audio peripherals to enable face tracking, arm movement, and voice dialogue.
- Contributed to **3D modelling and prototyping** of the robot enclosure using AutoCAD, **placing 3rd** in the UoA ECSE Design Competition.

Flappy Universe - VHDL | Team Project | 2026

- Implemented a **Flappy Bird-style game** in VHDL on an **Altera FPGA** with VGA signal generation, PS/2 mouse input, sprite rendering from ROM, and an LFSR random number generator.
- Developed a **layered graphics pipeline** using modular VHDL components and a pixel-priority compositing system to render animated game scenes at VGA resolution in real time.

Smart Energy Monitor - C & Embedded | Team Project | 2025

- Designed an embedded **energy monitoring system** using ATmega microcontrollers with sensor interfacing, ADC processing, and signal conditioning **to measure and display real-time energy usage**.
- Produced a validated PCB schematic using **Altium Designer and LTspice simulations** to achieve accurate sensor-to-display signal conditioning across the full hardware stack.

RoastWork Analytics - Python | Team Project | 2026

- Developed a **PyQt6 desktop analytics app** featuring multi-segment sales dashboards and a **scikit-learn** forecasting module using Regression and Holt's Exponential Smoothing, validated with MAE and RMSE metrics.

EXPERIENCE

Teaching Assistant @ University of Auckland

Jun 2026 - Present

- Delivered teaching support across tutorials, lab sessions, and test invigilation for **ELECTENG 292: Electronics** to strengthen **student leadership** and technical laboratory techniques.

Robotics Instructor @ Creative Imaginary Lab (ciLab)

Apr 2026 - Present

- Instructed students in robot **hardware assembly and software programming** to complete missions, while coaching and supporting teams in preparation for **nationwide robotics competitions**.

Front of House (FoH) @ Twelve Restaurant

Jul 2024 - Jan 2025

- Delivered exceptional customer service and collaborated with colleagues to maintain smooth front of house operations, efficiently resolving complex customer situations.

SKILLS

Languages & Frameworks: Python, Java, C, JavaScript, R, VHDL, [React.js](#), Pandas, Scikit-Learn

Tools & Platform: AWS (Lambda, Kinesis, DynamoDB, Comprehend), Git/GitHub, Android Studio, MATLAB, Altium Designer

LICENSES & CERTIFICATIONS

Amazon Web Services (AWS) Certified Cloud Practitioner

Apr 2026

- Demonstrated foundational knowledge of **AWS cloud concepts**, core services, and cloud architecture best practices.

Amazon Web Services (AWS) Certified AI Practitioner

May 2026

- Demonstrated knowledge of AI and machine learning concepts, **AWS AI/ML services**, and responsible AI practices.

ACTIVITIES & LEADERSHIP

Full time Volunteer of IEEE (Institute of Electrical and Electronics Engineers) @ NZRO 2025

Jul 2025

Academic Team Executive of KEB (Korean Engineering Body)

Jul 2024 - Present

- Tutored junior engineering students and assisted in planning academic events for the student community.